

(†) The rings should be regarded as commutative rings with identity.

Proposition 0.1. : *If N is a submodule of the A -module M , then*

$$S^{-1}(M/N) \cong S^{-1}M/S^{-1}N.$$

Proof: Consider the exact sequence

$$0 \longrightarrow N \xrightarrow{i} M \xrightarrow{\pi} M/N \longrightarrow 0$$

then by the the funtoriality of S^{-1} , we know that

$$0 \longrightarrow S^{-1}N \xrightarrow{S^{-1}i} S^{-1}M \xrightarrow{S^{-1}\pi} S^{-1}(M/N) \longrightarrow 0$$

is exact. Thus we get

$$S^{-1}(M/N) \cong S^{-1}M / \ker S^{-1}\pi \cong S^{-1}M / \text{Im} S^{-1}i \cong S^{-1}M / S^{-1}N. \quad \square$$

Proposition 0.2. : *If N and P are submodules of M , then*

- (1) $S^{-1}(N + P) = S^{-1}N + S^{-1}P$;
- (2) $S^{-1}(N \cap P) = S^{-1}N \cap S^{-1}P$.

Proof: (1) Straightforward verification ✓;

(2) Since $N \cap P \subseteq N$ and $N \cap P \subseteq P$, hence $S^{-1}(N \cap P) \subseteq S^{-1}N$, $S^{-1}(N \cap P) \subseteq S^{-1}P$ and therefore $S^{-1}(N \cap P) \subseteq S^{-1}N \cap S^{-1}P$;

let $\frac{n}{s} \in S^{-1}N \cap S^{-1}P$, $\exists \frac{p}{t} \in S^{-1}P$ such that $\frac{n}{s} = \frac{p}{t}$, i.e., $\exists u \in S$ such that $u(nt - ps) = 0$ and hence $unt = psu$, thus $unt \in N \cap P$, hence $\frac{unt}{snt} = \frac{n}{s} \in S^{-1}(N \cap P)$, thus $S^{-1}N \cap S^{-1}P \subseteq S^{-1}(N \cap P)$. $\square$

Local Property

Definition 0.1. : Let A be a ring, M is an A -module. A property P is said to be a **local property** if P holds true in A (or M) $\Leftrightarrow P$ holds true in $A_{\mathfrak{p}}$ (or $M_{\mathfrak{p}}$) for every prime ideal $\mathfrak{p}$ in A .

Proposition 0.3. : *Let A be a ring and M is an A -module, then TFAE (short for The Following Are Equivalent):*

- (1) $M = 0$;
- (2) $M_{\mathfrak{p}} = 0$ for every prime ideal $\mathfrak{p}$;
- (3) $M_{\mathfrak{m}} = 0$ for every maximal ideal $\mathfrak{m}$.

Proof: (1) $\Rightarrow$ (2): trivial ✓;

(2) $\Rightarrow$ (3): since every maximal ideal is a prime ideal $\checkmark$;

(3) $\Rightarrow$ (1): (prove by contradiction) Suppose $M \neq 0$, i.e., $\exists x \in M$ and $x \neq 0$. Hence we can know that $\text{Ann}(x) \neq A$, therefore $\text{Ann}(x)$ is a proper ideal of A , which implies that we can find a maximal ideal $\mathfrak{m}$ of A such that $\text{Ann}(x) \subseteq \mathfrak{m}$. Thus $\frac{x}{1} \in M_{\mathfrak{m}}$, i.e., $\exists u \in S = A \setminus \mathfrak{m}$ such that $ux = 0$, i.e., $u \in \text{Ann}(x)$, which is a contradiction! $\square$

Proposition 0.4. : Let A be a ring and M, N are two A -modules, then TFAE (short for The Following Are Equivalent):

- (1) the A -module homomorphism $f : M \rightarrow N$ is injective (resp. surjective);
- (2) $f_{\mathfrak{p}} \triangleq S^{-1}f : M_{\mathfrak{p}} \rightarrow N_{\mathfrak{p}}$ is injective (resp. surjective) for any prime ideal $\mathfrak{p}$;
- (3) $f_{\mathfrak{m}} \triangleq S^{-1}f : M_{\mathfrak{m}} \rightarrow N_{\mathfrak{m}}$ is injective (resp. surjective) for any maximal ideal $\mathfrak{m}$.

Proof: (1) $\Rightarrow$ (2) $\Rightarrow$ (3): $\checkmark$;

(3) $\Rightarrow$ (1): (a) Suppose $m \in \ker f$, hence $f(m) = 0$ and $f_{\mathfrak{m}}(\frac{m}{1}) = \frac{f(m)}{1} = \frac{0}{1} \in N_{\mathfrak{m}}$. By the injectivity of $f_{\mathfrak{m}}$, we can know that $\frac{m}{1} = \frac{0}{1}$, i.e., $\exists u \in S = A \setminus \mathfrak{m}$ ($\forall \mathfrak{m}$) such that $um = 0$, so $u \in \text{Ann}(m)$. If $m \neq 0$, then $\text{Ann}(m)$ is a proper ideal of A , then there exists a maximal ideal $\mathfrak{m}$ of A such that $\text{Ann}(m) \subseteq \mathfrak{m}$, hence $u \in \text{Ann}(m) \subseteq \mathfrak{m}$, which is a contradiction! So we have $m = 0$ and f is injective;

(b) By the condition (3), we know that $f_{\mathfrak{m}} : M_{\mathfrak{m}} \rightarrow N_{\mathfrak{m}}$ is surjective for $\forall \mathfrak{m}$. i.e., $\text{coker } f_{\mathfrak{m}} = 0$. Since

$$M \xrightarrow{f} N \longrightarrow \text{coker } f \longrightarrow 0$$

is exact, hence we have

$$M_{\mathfrak{m}} \xrightarrow{f_{\mathfrak{m}}} N_{\mathfrak{m}} \longrightarrow (\text{coker } f)_{\mathfrak{m}} \longrightarrow 0$$

is exact. Hence $(\text{coker } f)_{\mathfrak{m}} \cong \text{coker } f_{\mathfrak{m}} = 0$ for any maximal ideal $\mathfrak{m}$, thus by Proposition 0.3, we get $\text{coker } f = 0$, i.e., f is surjective. $\square$

Flatness Is A Local Property

Proposition 0.5. : Let M be an A -module, then TFAE:

- (1) M is a flat A -module;
- (2) $M_{\mathfrak{p}}$ is a flat $A_{\mathfrak{p}}$ -module for any prime ideal $\mathfrak{p}$;
- (3) $M_{\mathfrak{m}}$ is a flat $A_{\mathfrak{m}}$ -module for any maximal ideal $\mathfrak{m}$.

Proof: (1) $\Rightarrow$ (2): Suppose N and P are two $A_{\mathfrak{p}}$ -modules and suppose $f \in \text{Hom}_{A_{\mathfrak{p}}}(N, P)$ is injective. Then we can know that N, P are A -modules via $a \mapsto \frac{a}{1}$. For $\forall a \in A$ and $\forall n \in N$, we have

$$f(an) = f\left(\frac{a}{1}n\right) = \frac{a}{1}f(n) = af(n)$$

which shows that $f \in \text{Hom}_A(N, P)$ and it is injective naturally.

Since M is a flat A -module, so $f \otimes 1 \in \text{Hom}_A(N \otimes_A M, P \otimes_A M)$ is injective. By the functor S^{-1} , we know that $(f \otimes 1)_{\mathfrak{p}} \in \text{Hom}_{A_{\mathfrak{p}}}((N \otimes_A M)_{\mathfrak{p}}, (P \otimes_A M)_{\mathfrak{p}})$ is injective.

Since

$$\begin{aligned} N_{\mathfrak{p}} \otimes_{A_{\mathfrak{p}}} M_{\mathfrak{p}} &\cong (N \otimes_A M)_{\mathfrak{p}} \\ \phi_N : \frac{n}{t} \otimes \frac{m}{s} &\mapsto \frac{n \otimes m}{ts} \end{aligned}$$

and

$$\begin{aligned} P_{\mathfrak{p}} \otimes_{A_{\mathfrak{p}}} M_{\mathfrak{p}} &\cong (P \otimes_A M)_{\mathfrak{p}} \\ \phi_P : \frac{p}{t} \otimes \frac{m}{s} &\mapsto \frac{p \otimes m}{ts} \end{aligned}$$

then

$$\begin{aligned} (\phi_P \circ f_{\mathfrak{p}} \otimes 1)\left(\frac{n}{t} \otimes \frac{m}{s}\right) &= \phi_P\left(f_{\mathfrak{p}}\left(\frac{n}{t}\right) \otimes \frac{m}{s}\right) = \phi_P\left(\frac{f(n)}{t} \otimes \frac{m}{s}\right) = \frac{f(n) \otimes m}{ts} \\ ((f \otimes 1)_{\mathfrak{p}} \circ \phi_N)\left(\frac{n}{t} \otimes \frac{m}{s}\right) &= (f \otimes 1)_{\mathfrak{p}}\left(\frac{n \otimes m}{ts}\right) = \frac{(f \otimes 1)(n \otimes m)}{ts} = \frac{f(n) \otimes m}{ts} \end{aligned}$$

thus $\phi_P \circ f_{\mathfrak{p}} \otimes 1 = (f \otimes 1)_{\mathfrak{p}} \circ \phi_N$, i.e., we have the commutative diagram

$$\begin{array}{ccc} N_{\mathfrak{p}} \otimes_{A_{\mathfrak{p}}} M_{\mathfrak{p}} & \xrightarrow{f_{\mathfrak{p}} \otimes 1} & P_{\mathfrak{p}} \otimes_{A_{\mathfrak{p}}} M_{\mathfrak{p}} \\ \cong \downarrow & & \downarrow \cong \\ (N \otimes_A M)_{\mathfrak{p}} & \xrightarrow{(f \otimes 1)_{\mathfrak{p}}} & (P \otimes_A M)_{\mathfrak{p}} \end{array}$$

hence by the injectivity of $(f \otimes 1)_{\mathfrak{p}}$, we can get the injectivity of $f_{\mathfrak{p}} \otimes 1$, which means $M_{\mathfrak{p}}$ is a flat $A_{\mathfrak{p}}$ -module;

(2) $\Rightarrow$ (3): $\checkmark$;

(3) $\Rightarrow$ (1): Let

$$0 \longrightarrow N \xrightarrow{f} P$$

be exact. By the functor S^{-1} , we know

$$0 \longrightarrow N_{\mathfrak{m}} \xrightarrow{f_{\mathfrak{m}}} P_{\mathfrak{m}}$$

is exact. By condition (3), we know that

$$0 \longrightarrow N_{\mathfrak{m}} \otimes_{A_{\mathfrak{m}}} M_{\mathfrak{m}} \xrightarrow{f_{\mathfrak{m}} \otimes 1} P_{\mathfrak{m}} \otimes_{A_{\mathfrak{m}}} M_{\mathfrak{m}}$$

is exact for any maximal ideal $\mathfrak{m}$. Since we have the commutative diagram

$$\begin{array}{ccc} N_{\mathfrak{m}} \otimes_{A_{\mathfrak{m}}} M_{\mathfrak{m}} & \xrightarrow{f_{\mathfrak{m}} \otimes 1} & P_{\mathfrak{m}} \otimes_{A_{\mathfrak{m}}} M_{\mathfrak{m}} \\ \cong \downarrow & & \downarrow \cong \\ (N \otimes_A M)_{\mathfrak{m}} & \xrightarrow{(f \otimes 1)_{\mathfrak{m}}} & (P \otimes_A M)_{\mathfrak{m}} \end{array}$$

so we get the exact sequence

$$0 \longrightarrow (N \otimes_A M)_{\mathfrak{m}} \xrightarrow{(f \otimes 1)_{\mathfrak{m}}} (P \otimes_A M)_{\mathfrak{m}}$$

which implies $(f \otimes 1)_m$ is injective by Proposition 0.4 and hence M is a flat A -module. $\square$

We have known that proper ideals of $S^{-1}A$ is of the form $\{S^{-1}I \mid I \text{ is an ideal of } A, I \cap S = \emptyset\}$.

Now we talk about the prime ideals of $S^{-1}A$

Proposition 0.6. : *There is a one-to-one correspondence between the prime ideals of $S^{-1}A$ and the prime ideals of A that do not intersect with S :*

$$\mathfrak{p} \longleftrightarrow S^{-1}\mathfrak{p} \quad (\mathfrak{p} \cap S = \emptyset)$$

Proof: If $\mathfrak{p}$ is a prime ideal of A such that $\mathfrak{p} \cap S \neq \emptyset$, then $A/\mathfrak{p}$ is a field. Let $\bar{S}$ be the image of S under the natural homomorphism $A \rightarrow A/\mathfrak{p}$, then it's easy to see that $\bar{S}$ is a multiplicative set of $A/\mathfrak{p}$. Furthermore, $S^{-1}A/S^{-1}\mathfrak{p} \cong S^{-1}(A/\mathfrak{p}) \cong \bar{S}^{-1}(A/\mathfrak{p})$. Since $A/\mathfrak{p}$ is a field, so we know that $\bar{S}^{-1}(A/\mathfrak{p})$ is an integral domain. Thus $S^{-1}A/S^{-1}\mathfrak{p}$ is an integral domain and hence $S^{-1}\mathfrak{p}$ is a prime ideal of $S^{-1}A$;

Let P be a prime ideal of $S^{-1}A$, let $f : A \rightarrow S^{-1}A, a \mapsto \frac{a}{1}$. Denote $\mathfrak{p} \triangleq f^{-1}(P)$. Then we have $\mathfrak{p} \cap S = \emptyset$ and $P = S^{-1}\mathfrak{p}$. It's clear that $\mathfrak{p}$ is a prime ideal of A . Indeed, let $xy \in \mathfrak{p}$ with $x, y \in A$, then $xy \in f^{-1}(P)$ and $f(xy) = f(x)f(y) \in P$, which implies $f(x) \in P$ or $f(y) \in P$, i.e., $x \in \mathfrak{p}$ or $y \in \mathfrak{p}$. $\square$